

AMAZON REVIEWS SENTIMENT ANALYSIS USING NATURAL LANGUAGE PROCESSING (NLP)

Insight Report

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Program: AnalystLab Africa Data Science Internship Program

Project: Week 5 – Natural Language Processing (NLP)

Dataset: Amazon Reviews Dataset (Kaggle)

Tools: Python, Pandas, NLTK, Scikit-learn, TF-IDF, Logistic Regression, Matplotlib, Seaborn, WordCloud, Google Colab

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1.0 Executive Summary

Customer reviews represent a valuable source of unstructured data that can provide insights into customer experiences, preferences, and satisfaction levels. This project applied Natural Language Processing (NLP) and Machine Learning techniques to analyze Amazon customer reviews and classify sentiments as either positive or negative. Through text preprocessing, exploratory text analysis, TF-IDF vectorization, and Logistic Regression modeling, the study successfully transformed textual feedback into actionable insights. The developed model achieved an accuracy of 87.93%, demonstrating strong capability in identifying customer sentiment and highlighting the potential of NLP-driven solutions for customer feedback analysis.

2.0 Dataset Description

The dataset used for this project was the Amazon Reviews Dataset obtained from Kaggle. The dataset consists of customer-generated product reviews accompanied by sentiment labels in FastText format. Reviews labeled as "__label__1" represent negative sentiment, while reviews labeled as "__label__2" represent positive sentiment.

To ensure computational efficiency while maintaining representativeness, a sample of 50,000 reviews was extracted from the training dataset for analysis. The dataset contains diverse

customer opinions across various products, making it suitable for sentiment classification and text mining applications.

3.0 Methodology

The project followed a structured Natural Language Processing workflow consisting of data preparation, text analysis, feature engineering, model development, and evaluation.

The preprocessing stage involved converting all text to lowercase, removing punctuation, numbers, and special characters, and eliminating common stopwords. These steps reduced textual noise and improved the quality of the data used for modeling.

Following preprocessing, Exploratory Text Analysis (ETA) was conducted to identify frequently occurring words and recurring themes within customer reviews. Word frequency analysis and word cloud visualizations were employed to uncover dominant terms and patterns in customer feedback.

To prepare the textual data for machine learning, the cleaned reviews were transformed into numerical representations using Term Frequency–Inverse Document Frequency (TF-IDF) vectorization. TF-IDF was selected because it emphasizes informative words while reducing the influence of overly common terms that contribute little predictive value.

A Logistic Regression classifier was then trained on the TF-IDF features. The dataset was divided into training and testing subsets using an 80:20 ratio to evaluate the model's ability to generalize to unseen reviews.

4.0 Key Findings

The analysis revealed several notable patterns within customer feedback. Frequently occurring words included "book," "one," "movie," "love," and "good," suggesting that customers often discussed product experiences, entertainment-related items, and overall satisfaction.

The prevalence of positive expressions such as "love" and "good" indicates that a substantial portion of the sampled reviews reflected favorable customer experiences. The word cloud further reinforced this observation by highlighting positive and product-related terms as dominant features within the dataset.

The application of TF-IDF successfully distinguished meaningful sentiment-bearing words from commonly occurring terms. This improved the model's ability to recognize patterns associated with positive and negative reviews.

The findings demonstrate that customer reviews contain valuable business intelligence capable of revealing customer preferences, product perceptions, and overall satisfaction trends.

5.0 Model Performance

The Logistic Regression model demonstrated strong predictive performance across all evaluation metrics.

Table 1: Evaluation Results for the Logistic Regression Model

Metric	Score
Accuracy	87.93%
Precision	87.15%
Recall	89.24%
F1-Score	88.18%

The high accuracy indicates that the model correctly classified the majority of customer reviews. Precision results suggest that positive sentiment predictions were highly reliable, while the recall

score demonstrates the model's effectiveness in identifying actual positive reviews. The F1-score confirms a strong balance between precision and recall, indicating robust overall performance.

The confusion matrix further showed that most customer reviews were correctly classified, with relatively few misclassifications between positive and negative sentiments.

6.0 Recommendations

Organizations handling large volumes of customer feedback should consider implementing sentiment analysis systems to automate the monitoring of customer opinions and product perceptions. Such systems can significantly reduce the time and effort required to manually review customer feedback.

Future studies may explore more advanced Natural Language Processing techniques, including Word2Vec, GloVe, and transformer-based architectures such as BERT, to further improve sentiment classification performance.

Additional preprocessing techniques, particularly lemmatization and advanced feature engineering methods, may also enhance model accuracy and interpretability.

Businesses can leverage insights derived from customer reviews to identify recurring concerns, improve product quality, strengthen customer satisfaction strategies, and support data-driven decision-making processes.

7.0 Conclusion

This project successfully demonstrated the application of Natural Language Processing and Machine Learning techniques for sentiment analysis of Amazon customer reviews. By converting unstructured textual data into meaningful numerical representations and training a Logistic Regression classifier, the study achieved strong predictive performance and generated valuable insights into customer sentiment.

The results highlight the effectiveness of NLP as a tool for extracting actionable information from large-scale customer feedback datasets. As organizations continue to accumulate substantial volumes of textual data, sentiment analysis will remain an essential analytical approach for understanding customer experiences and supporting strategic business decisions.